

PERIPHERAL ARTERIAL DISEASE – LOWER EXTREMITY

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INTRODUCTION

Peripheral Arterial Disease (PAD) is an occlusive disease of arteries such as the aorta, iliac and lower limb (LL) arteries. The major cause is peripheral atherosclerosis, while a less common cause is thromboembolic disease. The objective definition of PAD is a resting Ankle Brachial Index (ABI) < 0.9.

RISK FACTORS

These are similar to the coronary risk factors:

- Diabetes mellitus (DM)
- Hypertension
- Hyperlipidaemia
- Smoking
- Hyperhomocysteinemia
- Age more than 70 years #
- Known atherosclerotic coronary, carotid or renal arterial disease

May consider age > 50 years as a risk factor if other risk factors are present, especially DM and smoking

Note: Patients with PAD have the same CVD mortality risk as patients who have had AMI / CVA.

CLINICAL PRESENTATION

Classic claudication is defined as muscle discomfort in the lower limbs that is reproduced (triggered) by exercise and relieved by rest within 10 minutes. Only 10 - 35% of patients with PAD have classic claudication; others have atypical leg pain (40 - 50%) or are asymptomatic (20 - 50%). Atypical symptoms can include buttock or thigh and not involving the calves, and not resolving within 10 minutes of rest.

CLASSIFICATION BASED ON CLINICAL SYMPTOMS OR SIGNS

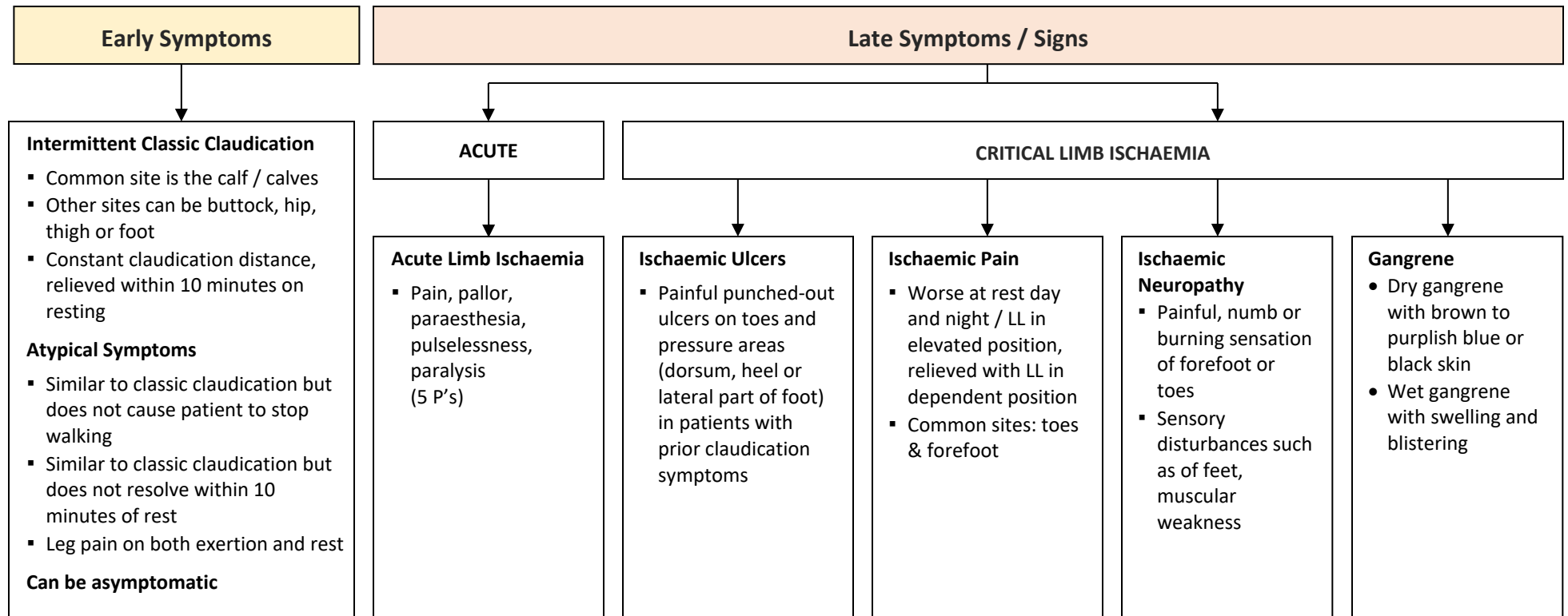
PVD (Fontaine's Classification)

- Stage 1** - Asymptomatic
- Stage 2** - Pain on walking
 - 2A** - Mild claudication - Distance more than 200 metres
 - 2B** - Moderate to severe claudication - Distance less than 200 metres
- Stage 3** - Ischaemic rest pain
- Stage 4** - Ulceration or gangrene

Stage 3 and 4 represent the most severe manifestations of PVD and are termed as Critical Limb Ischaemia (CLI).

HISTORY

Assess for risk factors and past medical history.



PHYSICAL EXAMINATION

Inspect:

- Feet for colour, temperature, integrity of skin and ulcers

Palpate:

- Peripheral pulses (femoral, popliteal, posterior tibial, dorsalis pedis) & grade
- Abdomen for aortic pulsation

Auscultate for bruit:

- Aortic, renal and femoral bruits
- Note: severely stenotic vessel → no bruit

Other suggestive lower extremity physical findings:

- Elevation pallor
- Dependent rubor

ASSESSMENT

Assessment is done by performing a resting Ankle-Brachial Index (ABI). ABI should be considered for the following:

1. Symptomatic patients. The ABI should be measured in both legs to confirm the diagnosis of lower extremity PAD and establish a baseline.
2. Patients with abnormal examination, e.g. poor capillary refill (> 3 seconds), weak pulses or ischaemic-appearing foot or non-healing wounds of the legs or feet.

MEASUREMENT OF ANKLE-BRACHIAL INDEX (ABI)

- Place BP cuff over upper arm.
- Inflate to above systolic BP.
- Place Doppler probe over brachial artery.
- Detect resumption of blood flow.
- If reading differs for both arms, take the higher reading.
- Similarly obtain reading for both ankles (take the higher value) by placing Doppler probe over posterior tibial & dorsalis pedis arteries.
- The highest of the two readings on one leg is divided by the higher of the two brachial measurements to obtain the ABI for that leg. So there is ABI for left and ABI for right.

ABI Grading of Severity of Peripheral Arterial Disease

ABI	Severity
0.90 or less	Abnormal
0.91 - 0.99	Borderline
1.00 - 1.40	Normal
Greater than 1.40	Non-compressible measurement

When ABI is greater than 1.40, Toe-brachial index (TBI) is measured to diagnose patients with suspected PAD. TBI is calculated by dividing the toe systolic pressure by the highest brachial systolic pressure.

TBI	Interpretation
≥ 0.7	Normal
0.5 to 0.69	Peripheral Arterial Disease
0.49 and below Or Toe Pressure < 30 mmHg	Severe peripheral arterial disease, need urgent referral

Note on Exercise ABI

The role of exercise ABI may be complicated and limited in Primary Care. These patients should be referred to Tertiary Care.

1. An exercise ABI measurement can be useful to diagnose lower extremity PAD in individuals who are at risk for lower extremity PAD who have a normal ABI (0.91 to 1.40), are without classic claudication symptoms and have no other clinical evidence of atherosclerosis. (It is usually performed when ABI falls in the normal range, yet clinical suspicion is still high whether it's classic claudication or atypical symptoms). However, a formal ABI testing in a vascular laboratory with waveform analysis can also help us look beyond just a number. Waveform, say triphasic (normal) to monophasic (abnormal) is very suggestive of disease.
2. The test is stopped when the patient is unable to walk further because of pain, defining maximal walking distance (WD).
3. A post-exercise ankle SBP decrease >30mmHg or a post-exercise ABI decrease >20% are diagnostic of lower extremity artery disease.

Note on ABI and TBI in NUP

1. ABI and TBI are not conducted for all patients with diabetes as part of diabetic foot screening (DFS) in NUP.
2. Toe pressure is incorporated into DFS and is only performed when the pedal pulses not palpable. Toe pressure is not done as baseline or annually.
3. ABI is performed when pedal pulses are absent and it is not possible to obtain the toe pressure (e.g. presence of 1st ray amputation) as the toe pressure is only done on the big toe.

DIFFERENTIAL DIAGNOSES

Other vascular causes of lower limb ischaemia:

- Arterial embolism
- Dissection
- Trauma
- Thrombosis of aneurysm
- Buerger's disease (thromboangiitis obliterans)

Non-vascular causes of “pseudoclaudication” (variable claudication distance, pain not relieved on standing alone, usually require sitting, taking weight off limb or changing body position)

- Spinal stenosis
- OA hip or knee
- Lumbar radiculopathy
- Nocturnal leg cramp
- Diabetic neuropathic pain
- Venous claudication

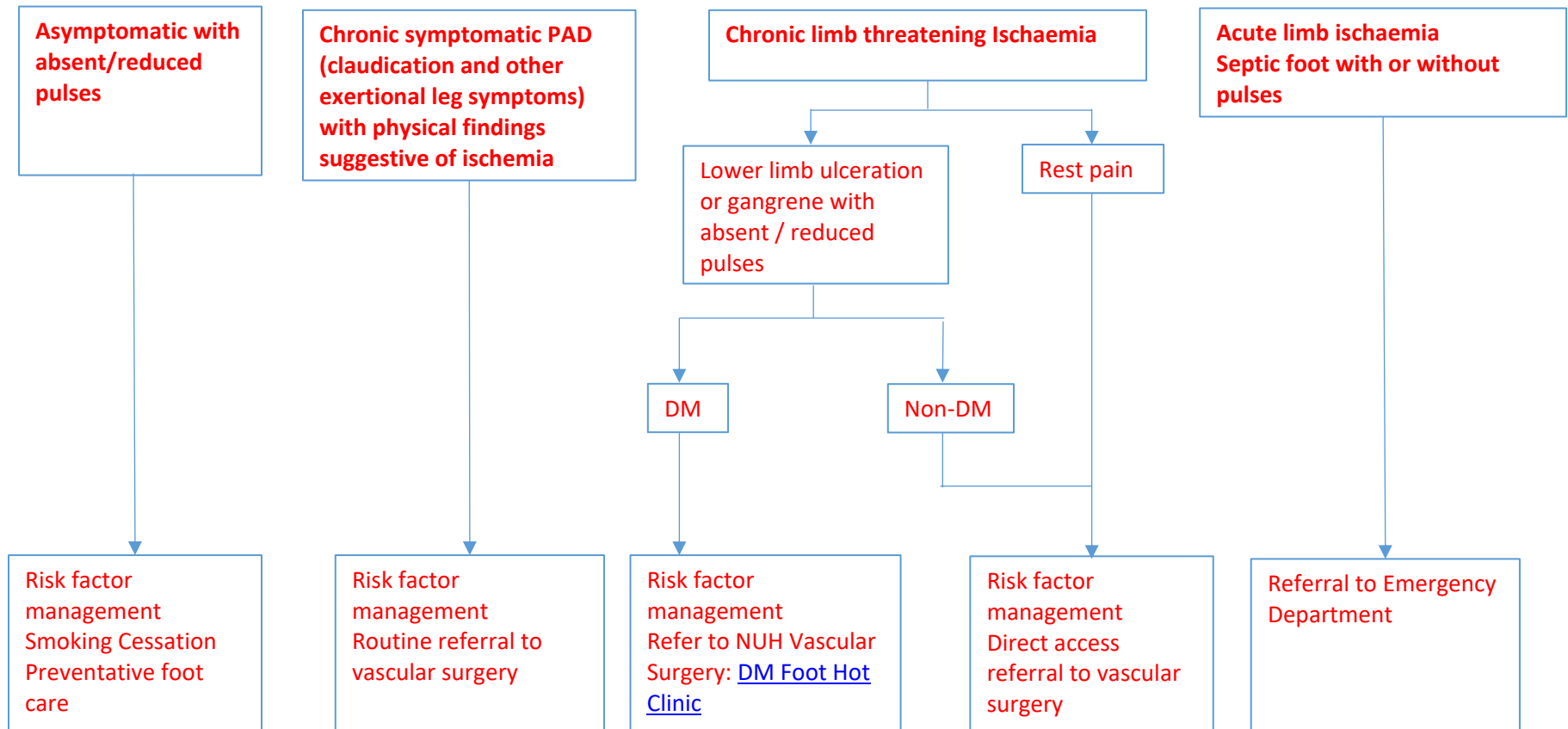
Note: PAD may co-exist with other disease conditions with similar presentation.

MANAGEMENT: 3-PRONGED APPROACH

Improve functional status (symptoms, quality of life and exercise capacity)	Preserve limb - decrease need for revascularisation procedure	Prevent progression of atherosclerosis & reduce cardiovascular morbidity / mortality
Supervised exercise programme in hospitals Cilostazol: is an effective therapy to improve symptoms & increase walking distance in patients with claudication. However, side effects include headache, diarrhoea, dizziness and palpitations. In 1 trial, 20% of patients discontinued cilostazol within 3 months. Pentoxifylline: is not effective for treatment of claudication	Regular surveillance Healthy foot behaviours • daily self-foot examination • wearing of shoes & socks • avoidance of barefoot walking • proper footwear Timely referral for revascularisation procedure where needed	<u>Control Risk Factors</u> Smoking Cessation DM: target HbA1c < 7% as appropriate. Hypertension: target <130/80mmHg * Beta-blockers are not contraindicated Hyperlipidaemia: LDL < 1.8 mmol/L or decrease it by ≥ 50% if baseline values are 1.8-3.5 mmol/L. Statins are recommended in all patients with PADs. <u>Drug Therapy</u> Anti-platelet: Aspirin 100 mg OM (Clopidogrel 75 mg OM, if aspirin cannot be used). Can be considered in asymptomatic patients with abnormal ABI (but low evidence level) ACE-I/ARB: recommended to reduce the risk of major adverse cardiac events in patients with PAD and hypertension SGLT2 inhibitors (empagliflozin, dapagliflozin, canagliflozin) and GLP1 Agonists (liraglutide and semaglutide): Effective at reducing major adverse cardiac events in patients with PAD and DM

INDICATIONS FOR REFERRAL TO HOSPITAL FOR FURTHER INVESTIGATION & MANAGEMENT

Suspected Peripheral Artery Disease Referral Workflow



REFERENCE

1. 2024 ACC/AHA/AACVPR/APMA/ABC/SCAI/SVM/SVN/SVS/SIR/VESS Guideline for the Management of Lower Extremity Peripheral Artery Disease: A Report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines
2. 2017 ESC Guidelines on the Diagnosis and Treatment of Peripheral Arterial Diseases, in collaboration with the European Society for Vascular Surgery (ESVS)